Physics-Informed Neural Networks on Darcy Flow below a Water Reservoir

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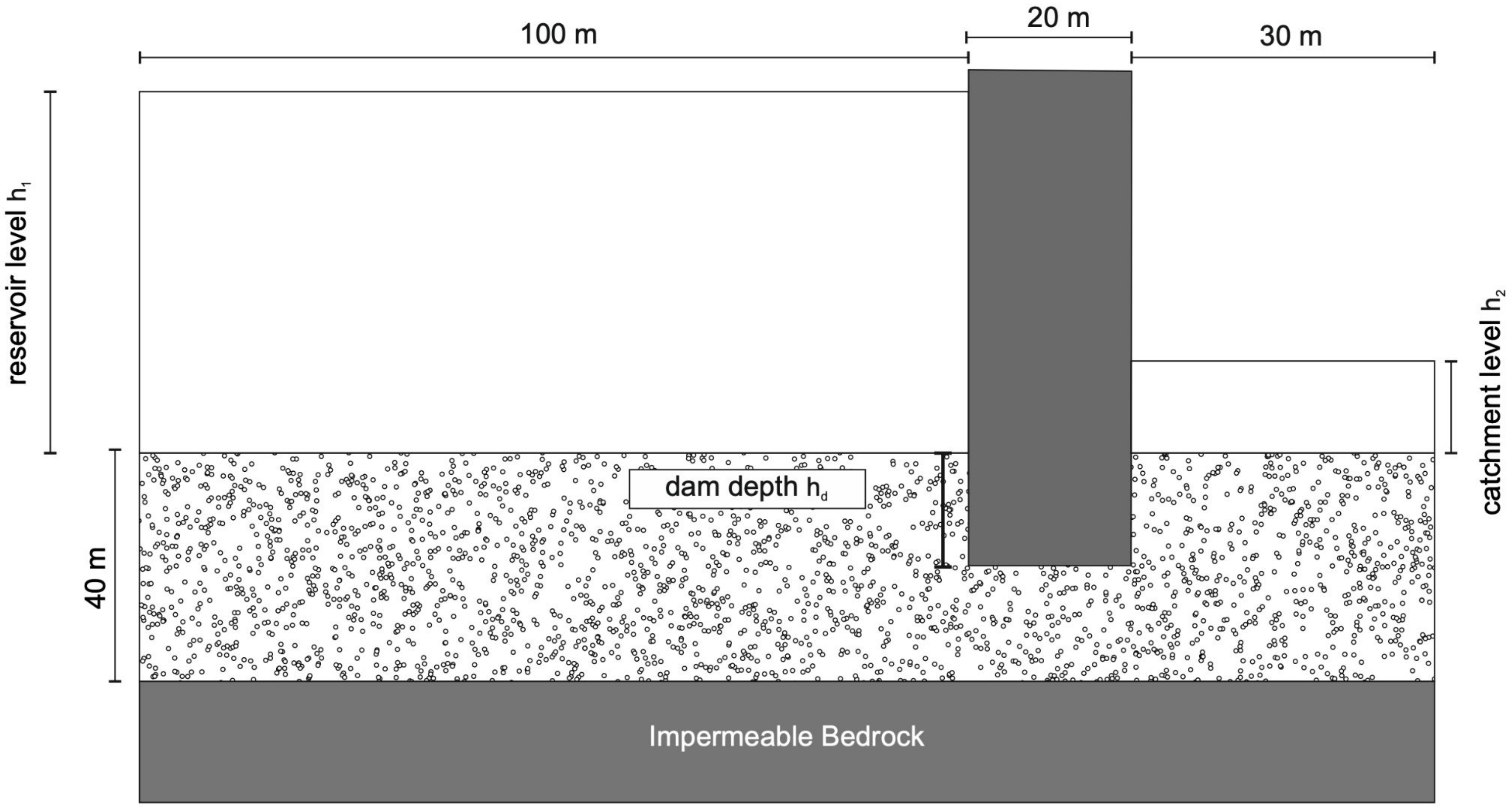
Overview

- Background
- Task 1: Non-dimensionalize the governing equation
- Task 2: Development of a forward PINN model
- Task 3: Development of an inverse PINN model
- Conclusions

General Overview of the Problem

Darcy Flow below a Water Reservoir

Overview of the Problem

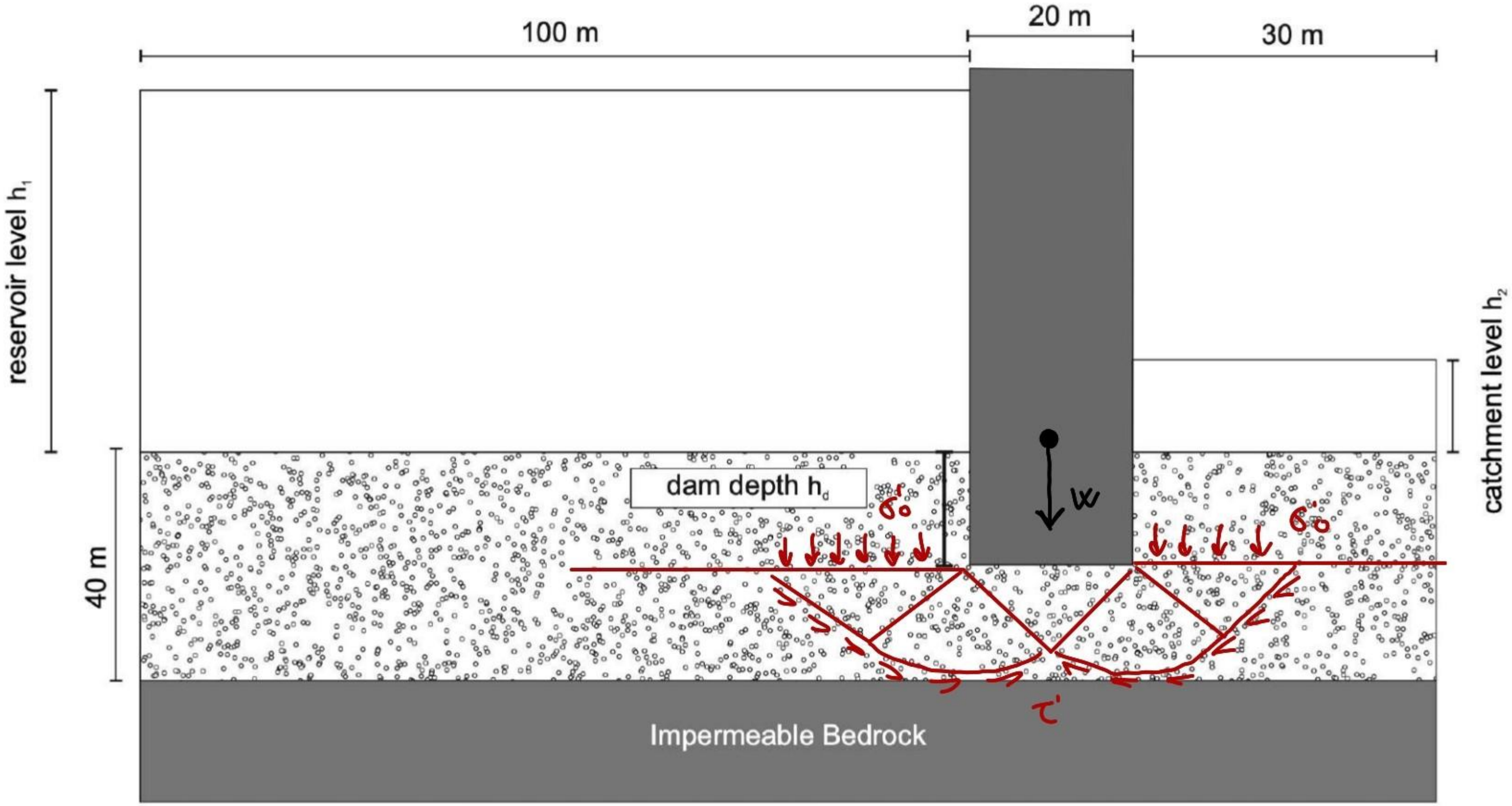


Inputs:
- Geometry
- Water levels

Conditions:
- Darcy's Law
- Principle of continuity

Output:
- Head (h)
- Hydraulic gradient
- Velocities and discharge
- (Permeability)

Importance of the Problem: hydraulic Effective stresses / bearing capacity

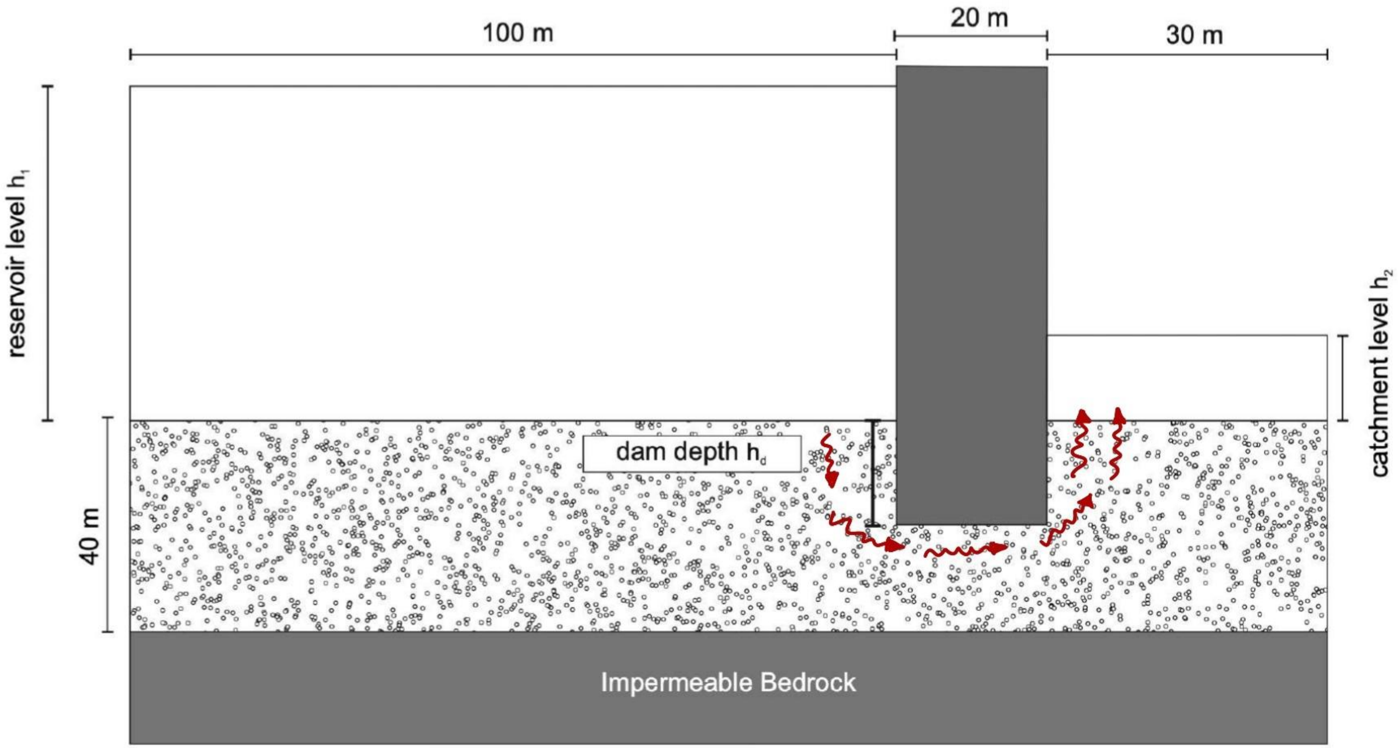


$$\tau = c + \sigma' \tan\phi$$

$$\sigma' = \sigma - u$$

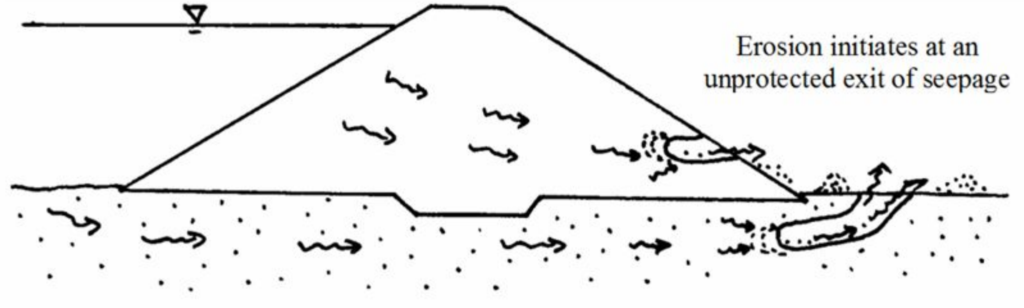
$$u = (h - z)\gamma_w$$

Importance of the Problem: hydraulic gradient



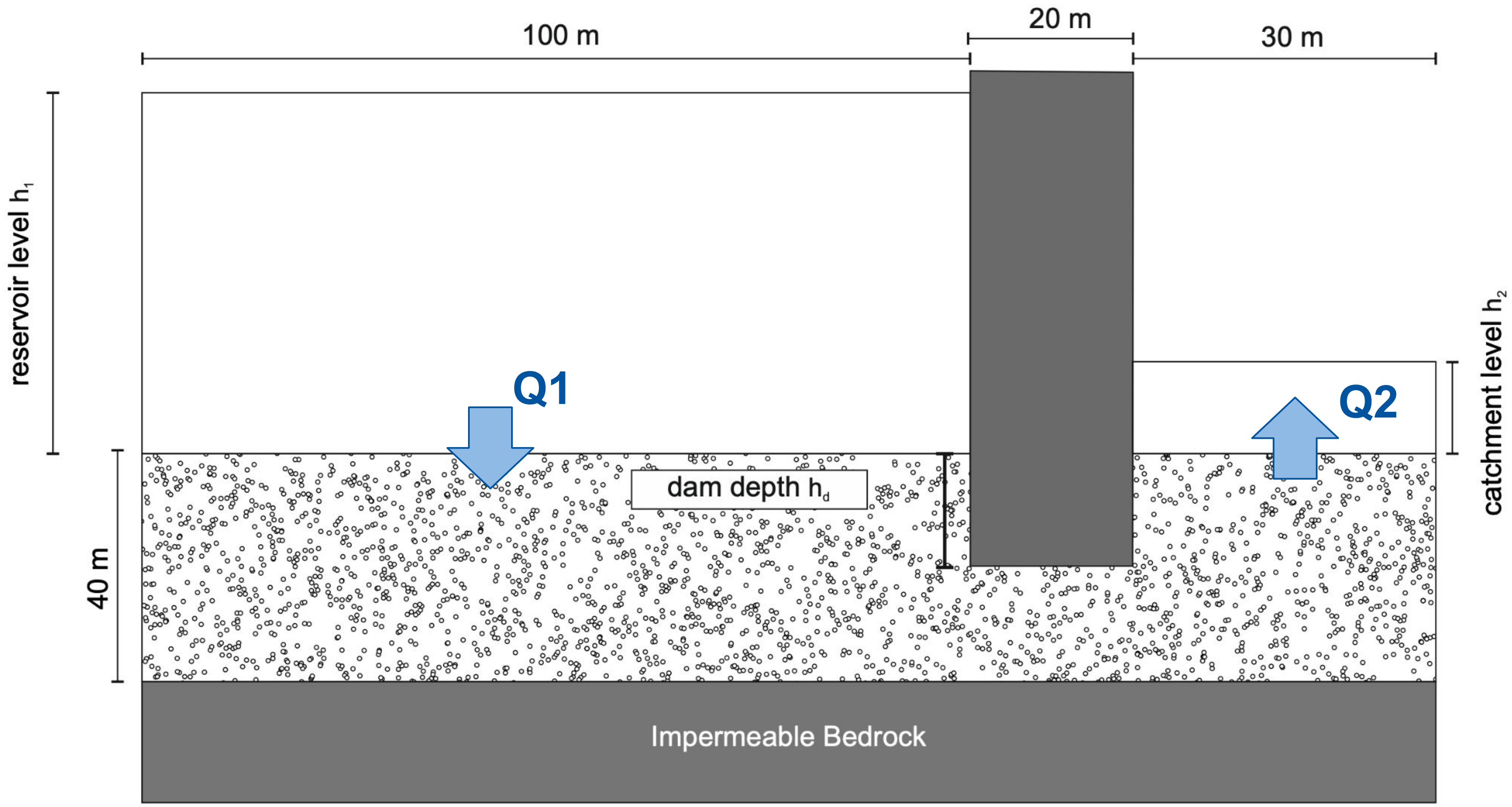
$$i = \nabla h$$

$$i_{crit} = \frac{\gamma_{sub}}{\gamma_s} = \frac{\gamma_s - \gamma_w}{\gamma_w}$$



source: Foster (1999)

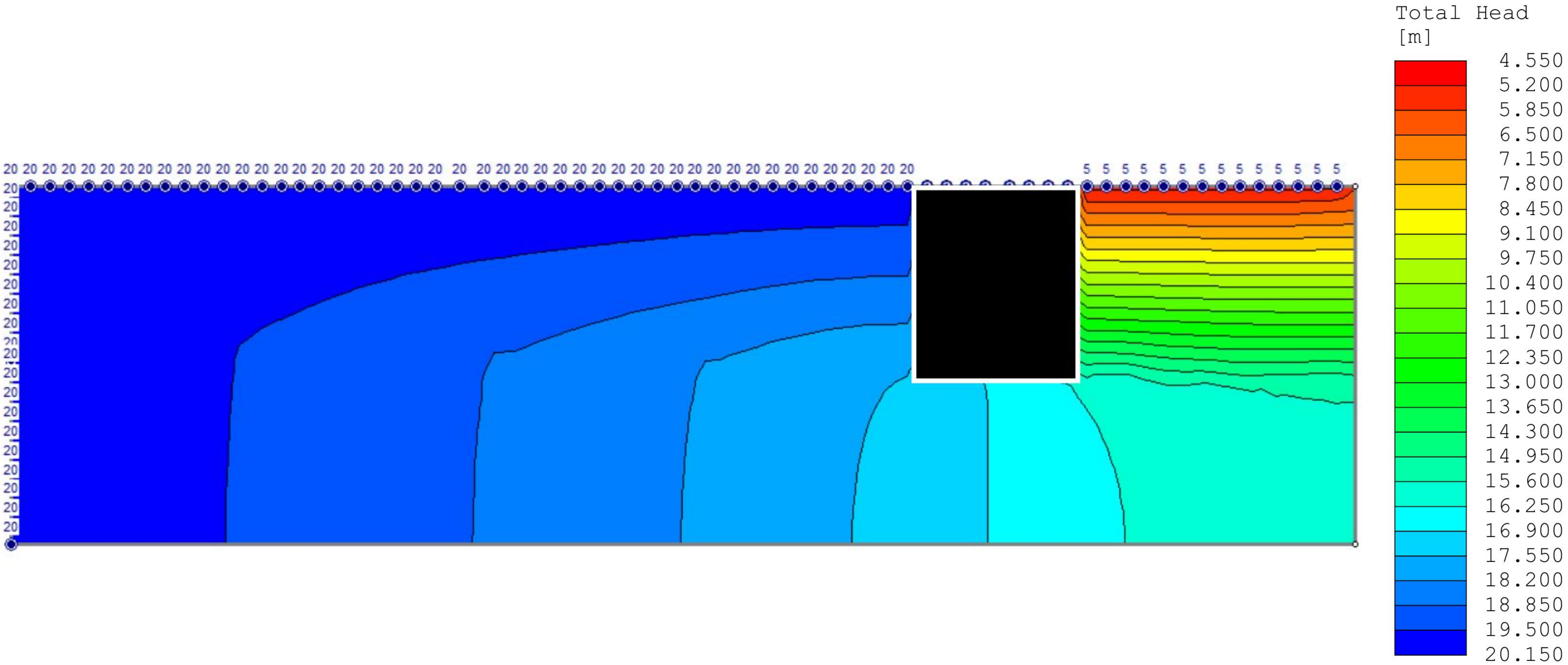
Importance of the Problem: Discharge



source:https://www.utilities-me.com
/water/article-5309-why-hydropower
-is-preferred-over-solar-and-wind

Q is measurable

FEM results (Slide2)



Task 1

Non-dimensionalization of the governing equation

Dimension Equations

Governing equation

$$\Delta h = \frac{\partial^2 h}{\partial x^2} + \frac{\partial^2 h}{\partial y^2} = 0$$

Dimensionless parameters

$$\hat{x} = \frac{x}{L}$$

$$\hat{y} = \frac{y}{L}$$

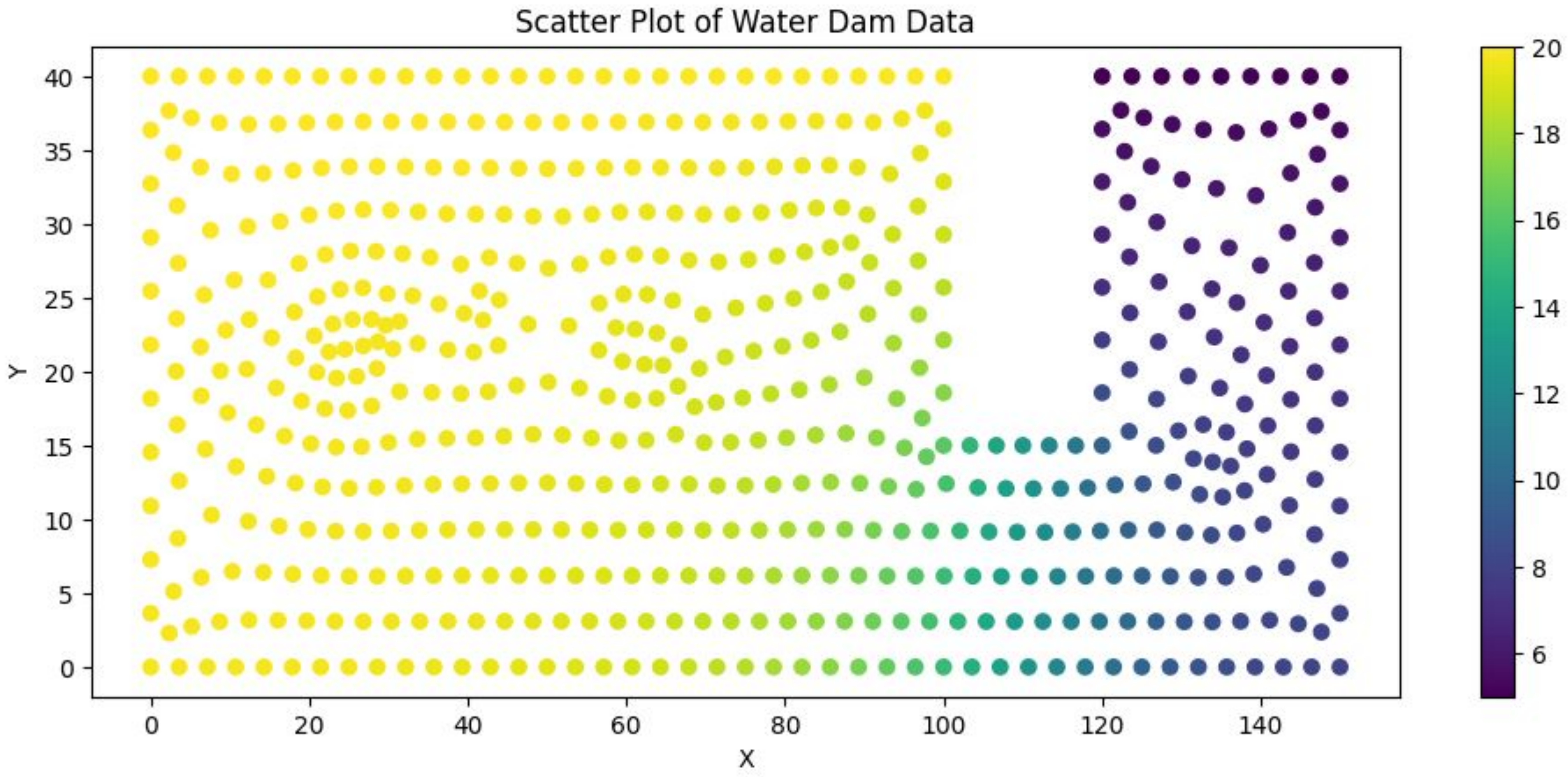
$$H = \frac{h - h_2}{h_1 - h_2}$$

Dimensionless governing equation and output

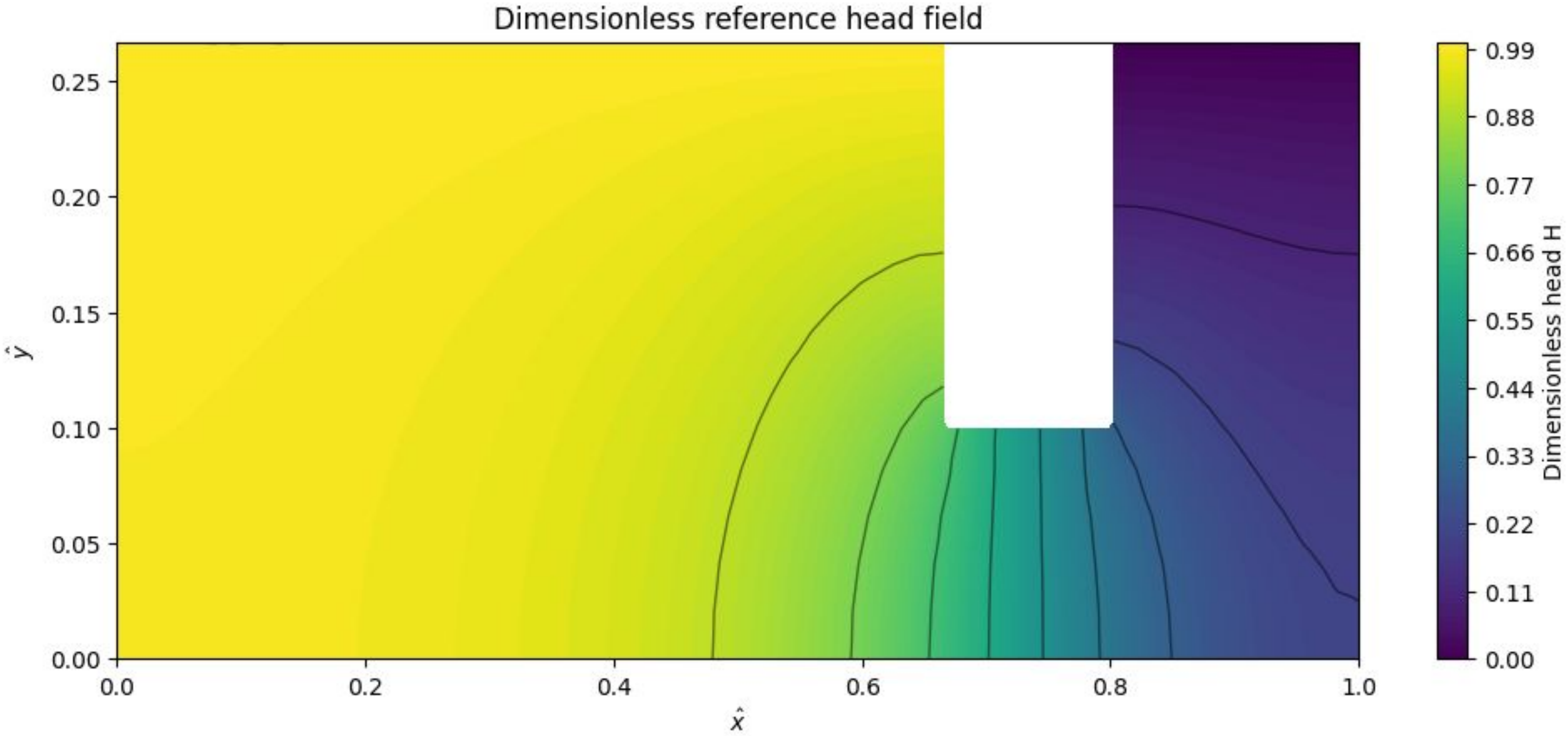
$$\frac{\partial^2 H}{\partial \hat{x}^2} + \frac{\partial^2 H}{\partial \hat{y}^2} = 0$$

$$h = h_2 + (h_1 - h_2)H$$

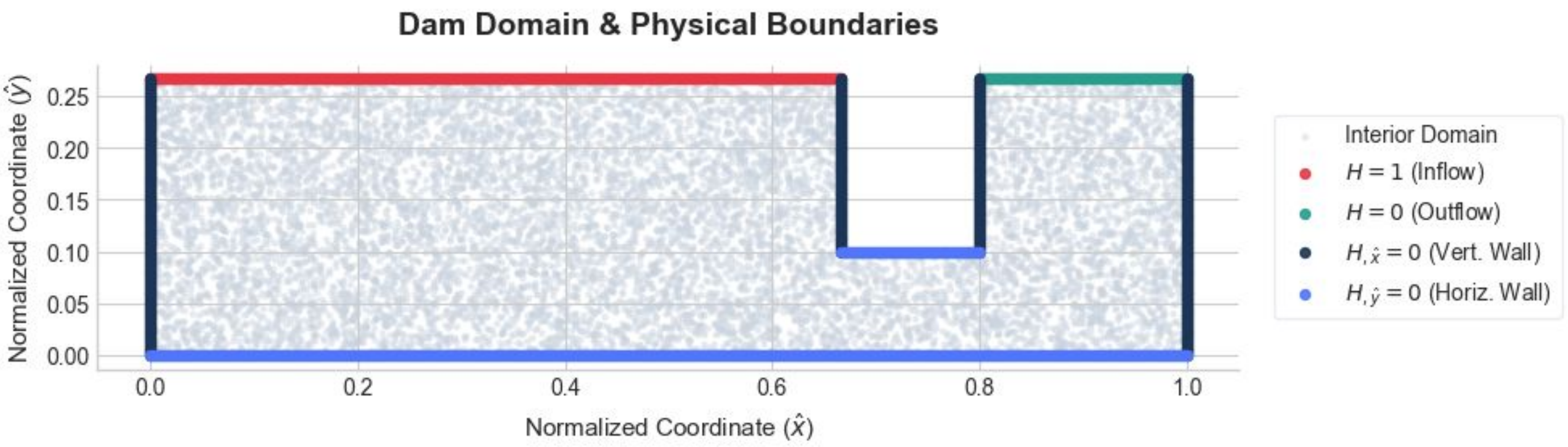
Scatter Plot of Water Dam in function of velocity (From Blue to Yellow)



Non-Dimensionalization



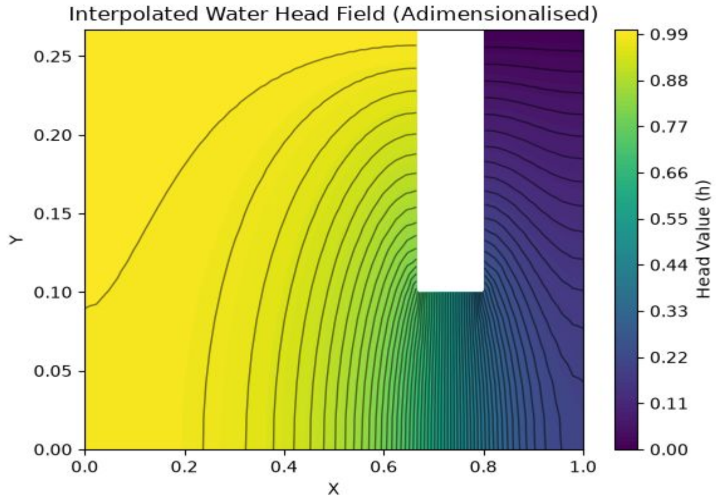
Visualized Boundary Conditions

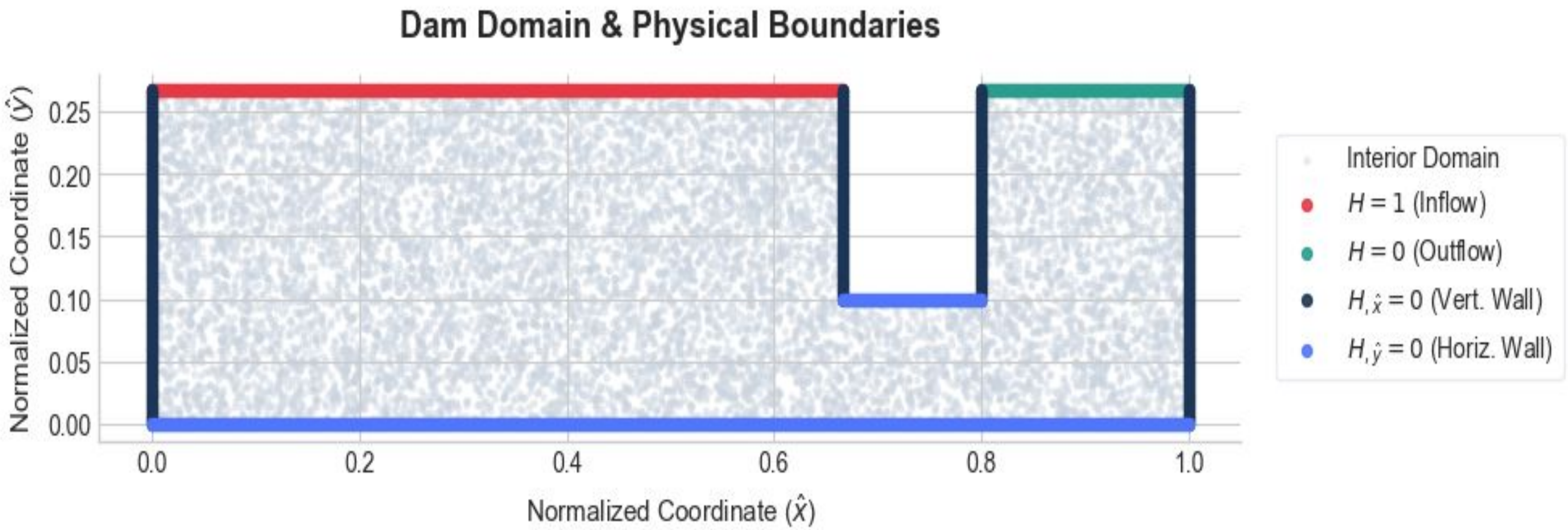


Task 2

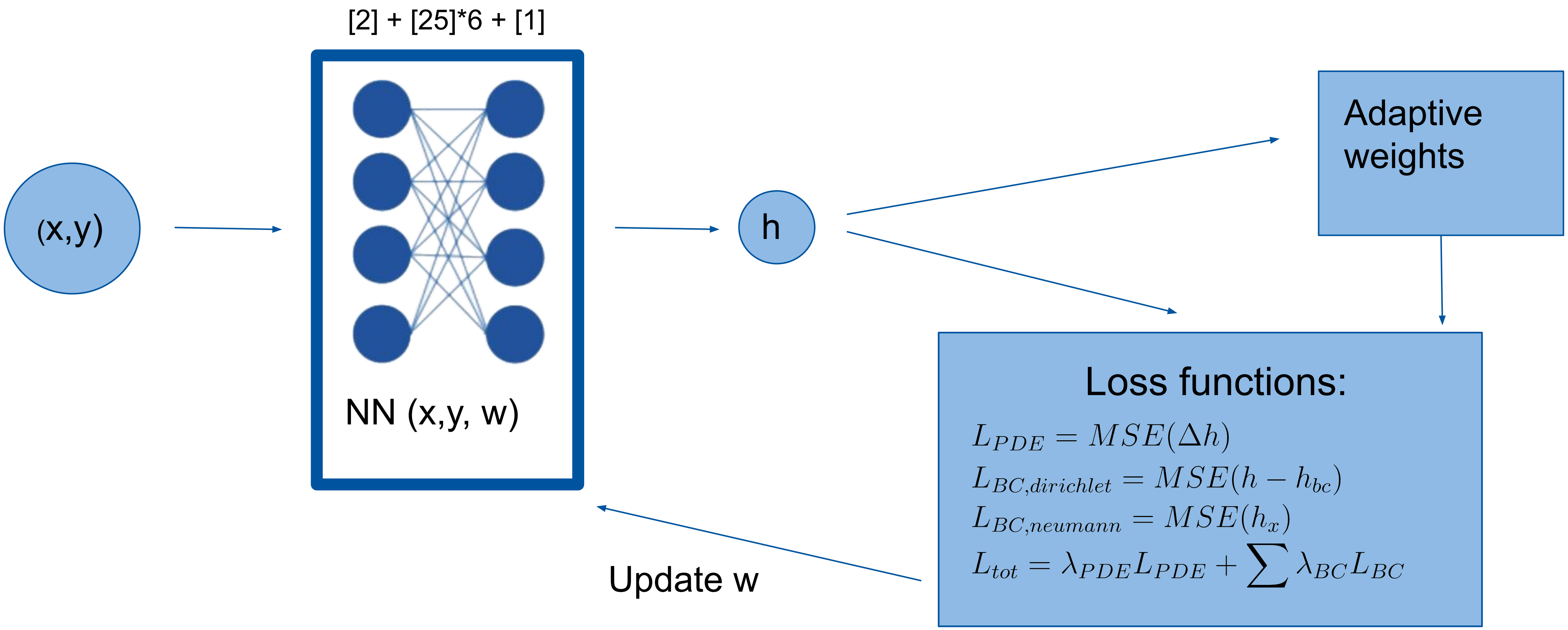
Train a forward Physics-Informed Neural Network (PINN) model that can estimate the pressure head distribution without using data.

Visualized Boundary Conditions





Task 2 Model

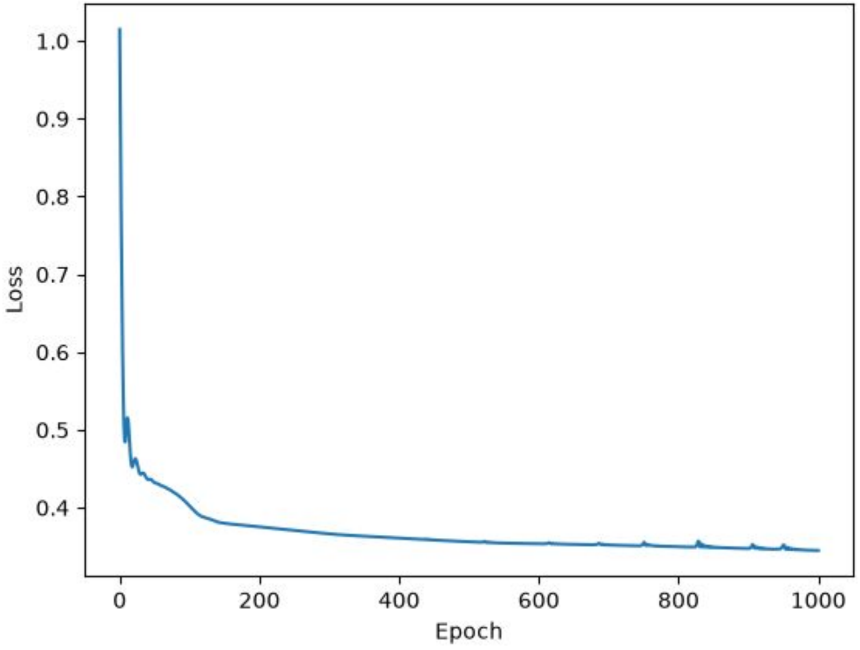


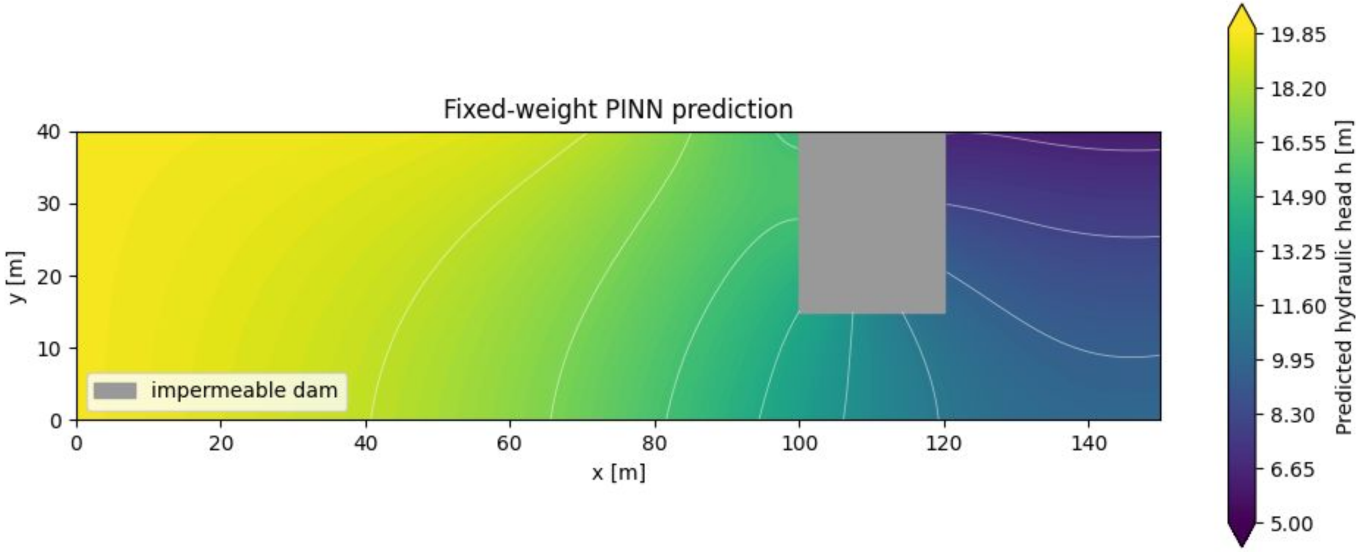
[2] + [25]*6 + [1]
(x,y)
NN (x,y, w)
h
Adaptive weights
Loss functions:
$L_{PDE} = MSE(\Delta h)$
$L_{BC,dirichlet} = MSE(h - h_{bc})$
$L_{BC,neumann} = MSE(h_x)$
$L_{tot} = \lambda_{PDE} L_{PDE} + \sum \lambda_{BC} L_{BC}$
Update w

Model 1: Fixed Weights

First approach

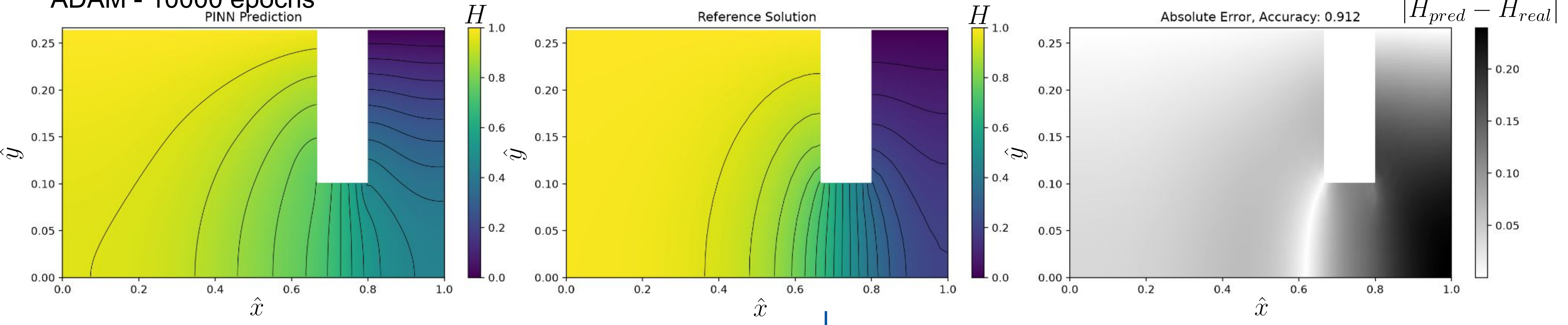
- Non-adaptive weights, loss hits plateau



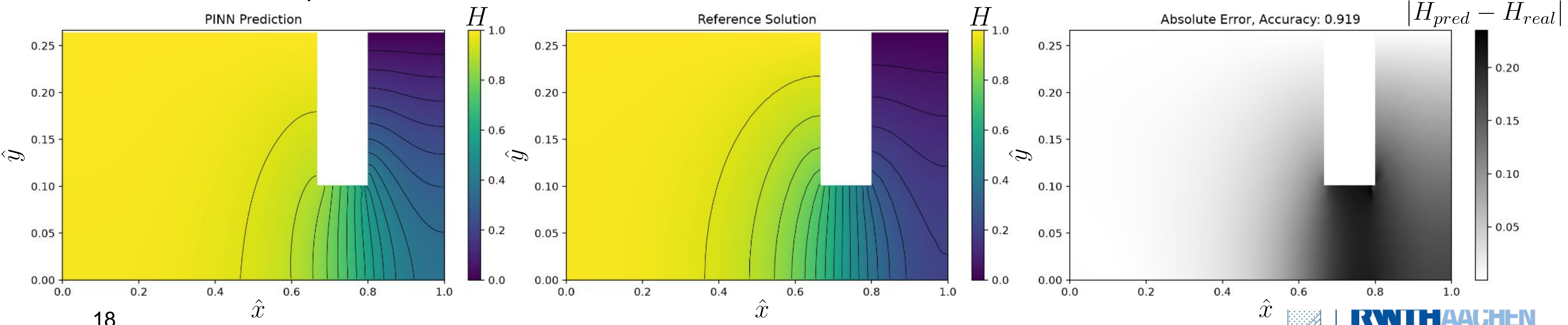


Model 2: NKTS adaptive weights

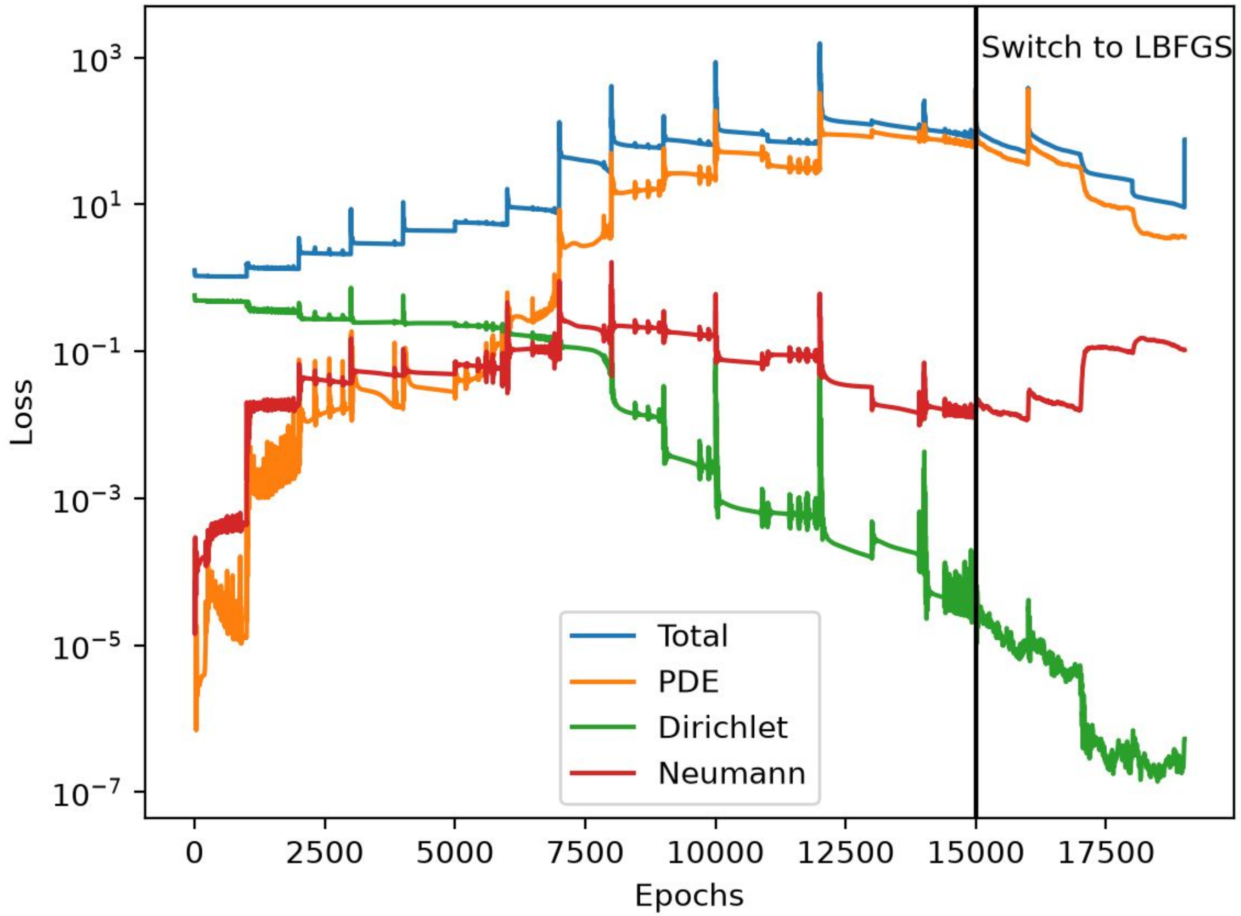
ADAM - 10000 epochs



LBFGS - 4005 extra epochs



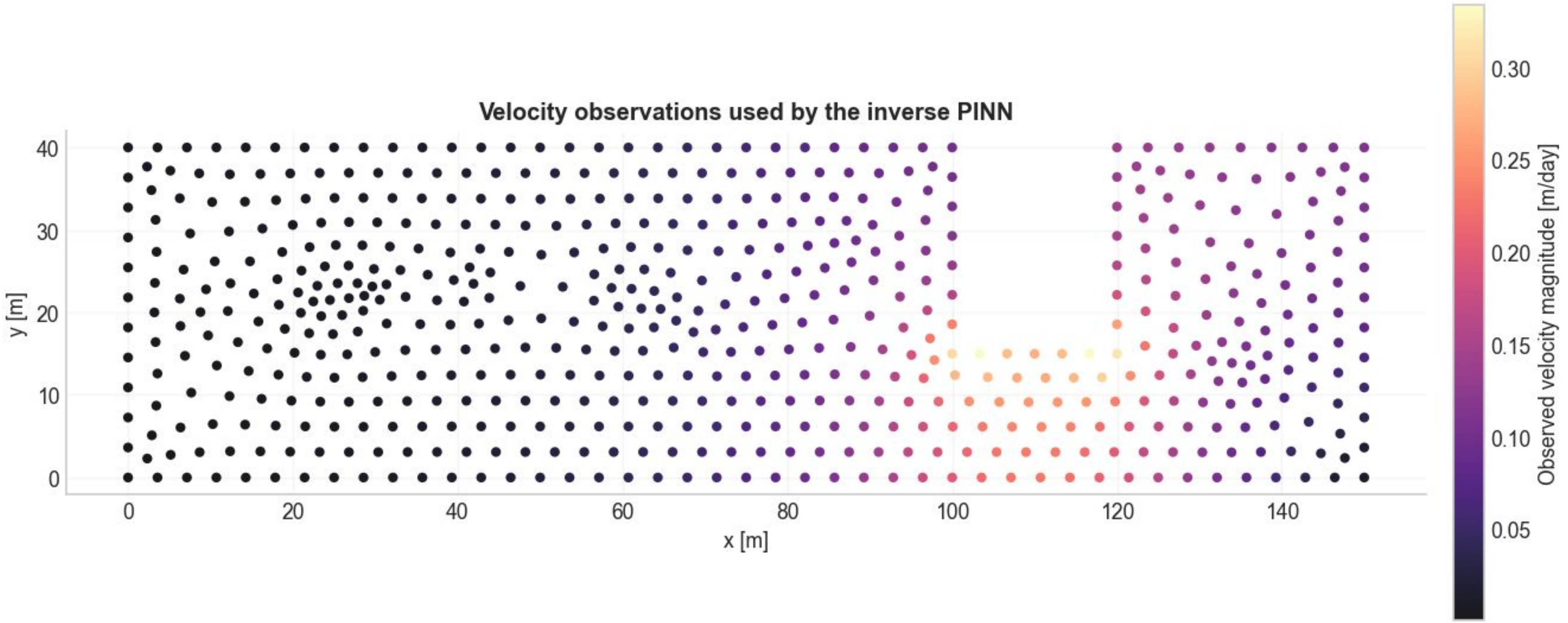
Model 2: NKTS adaptive weights



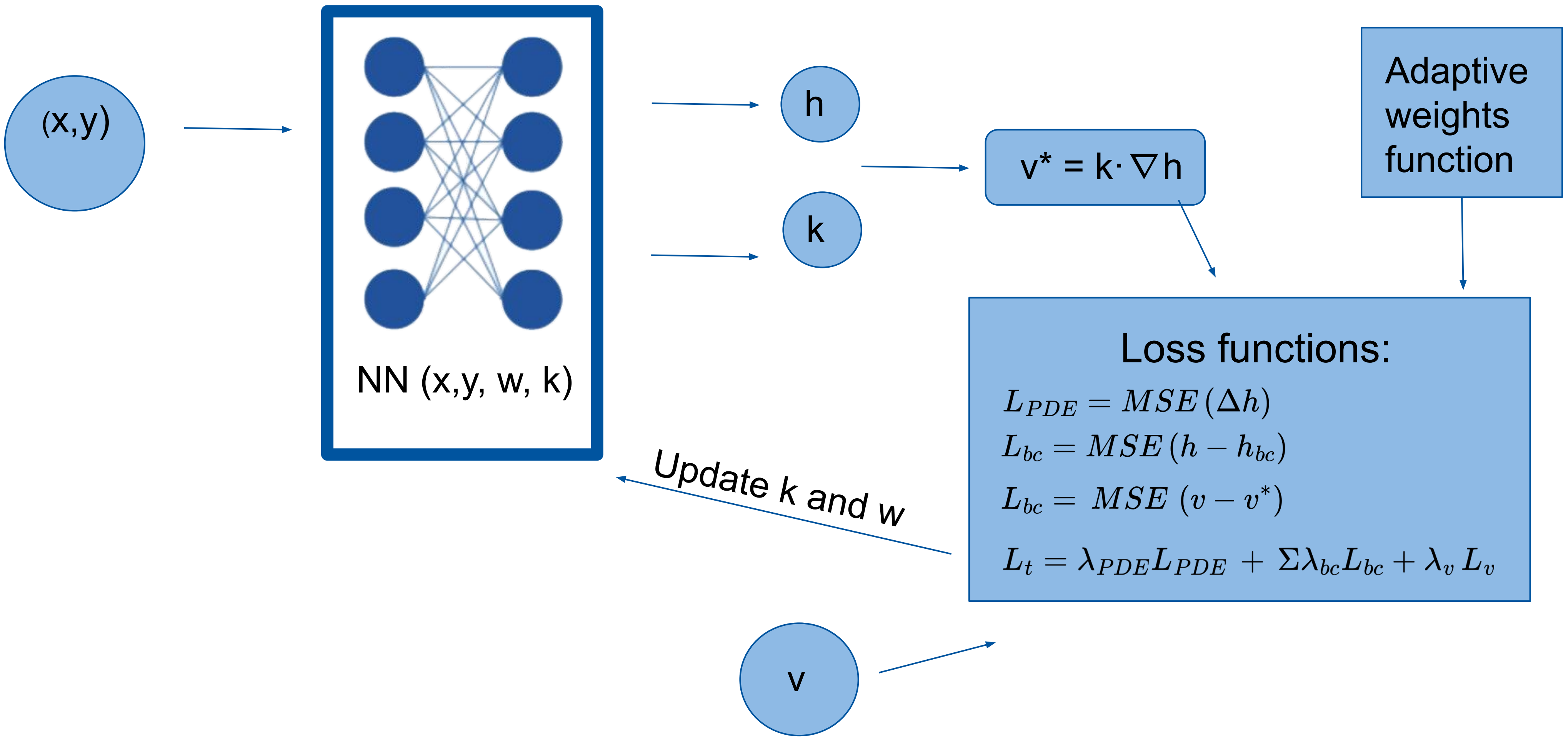
Task 3 - Development of a inverse PINN model

Training an inverse PINN model to estimate the hydraulic conductivity using the observed velocity fields. In this model, treat the hydraulic conductivity parameter k as a learnable quantity within the neural network.

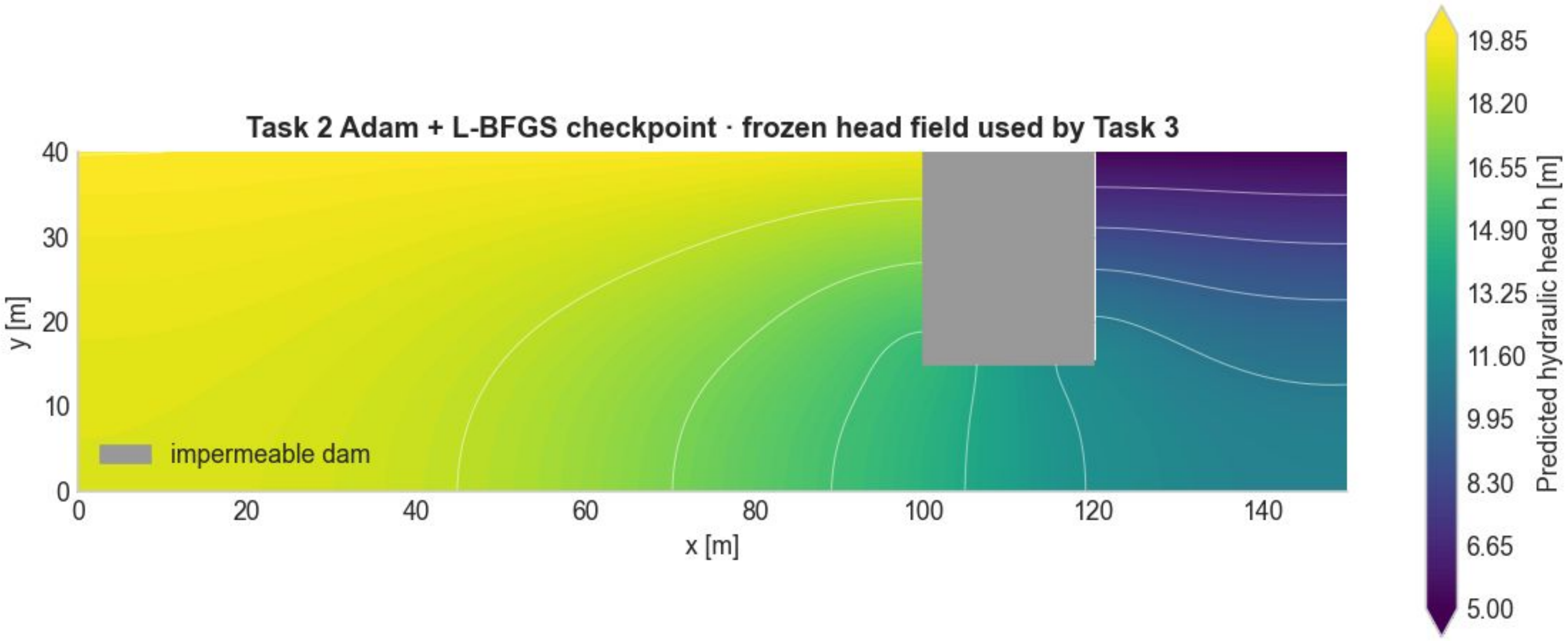
Task 3: Observed Velocity



Task 3: Model Description



Task 3: Reconstructing our Task 2, using results as initial weights of model



Task 3: Permeability and Velocity Approximation from predicted pressure heads

Our observed-velocity boundary/data loss is

$$\boxed{\mathcal{L}_v = \mathrm{MSE}(v_\theta, v^*) = \frac{1}{N}\sum_{i=1}^{N}\left(k|\nabla h_\theta(x_i, y_i)| - v_{real}\right)^2}.$$

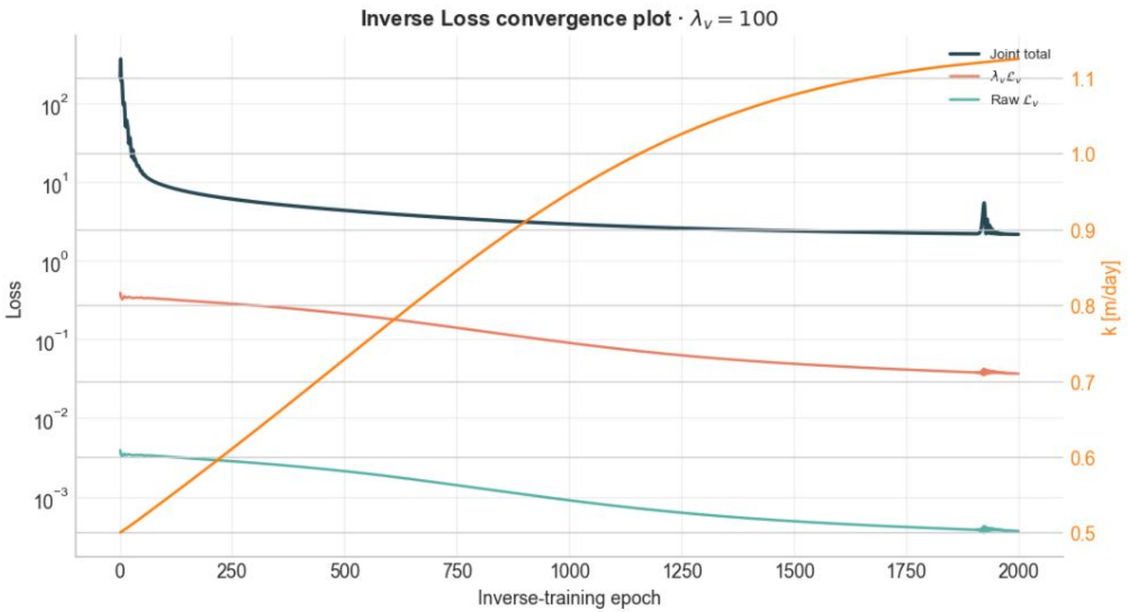
Task 3 adds this term to the original Task 2 losses:

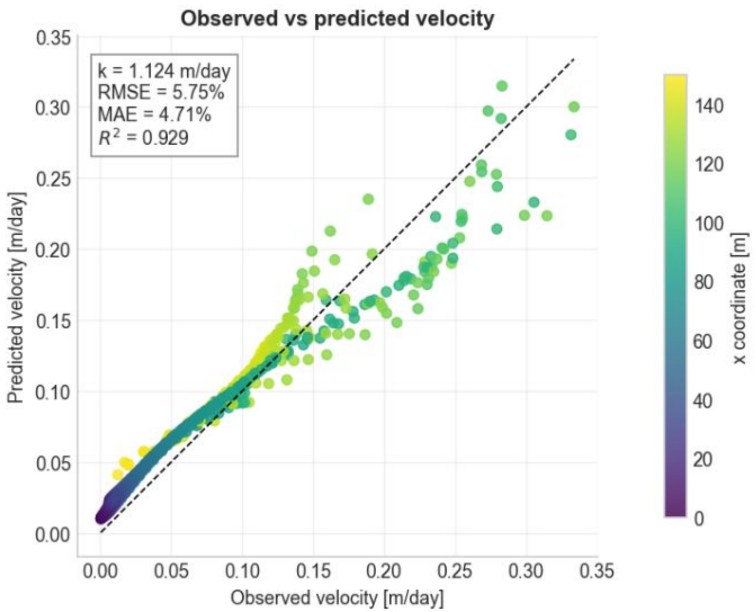
$$\boxed{\mathcal{L}_{inverse} = \mathcal{L}_{PDE} + \lambda_D\mathcal{L}_D + \lambda_N\mathcal{L}_N + \lambda_v\mathcal{L}_v}.$$

A deliberately large λ_v makes agreement with observed speed influential relative to the PDE and boundary terms. Adam updates the head-network weights and k_{raw} together, so `k` is learned inside one joint inverse model rather than calculated in post-processing.

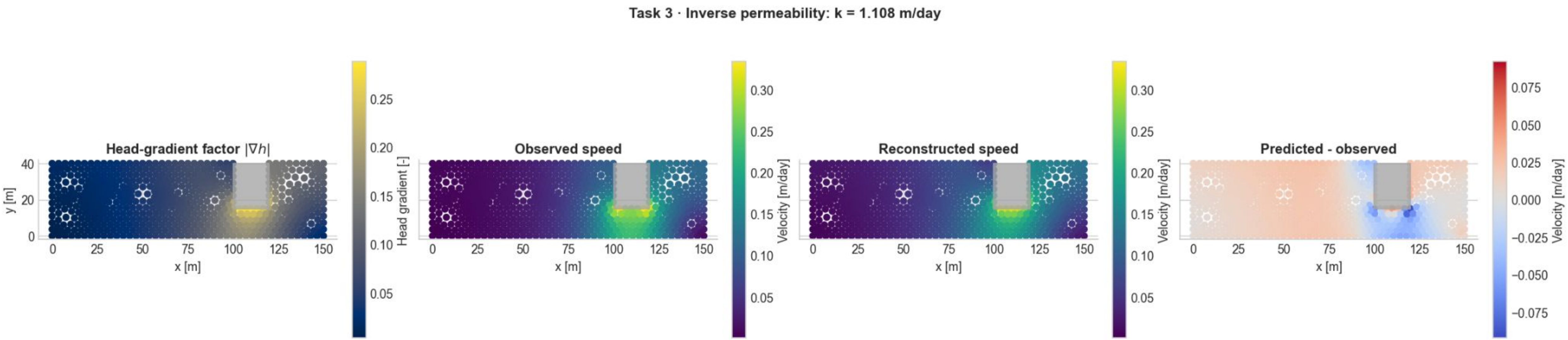
$$\text{Darcy's law:} \quad v_d = k_f * \nabla h$$

Task 3: Loss functions





Task 3: Velocity Approximation from predicted pressure heads



Normalized Velocity RMSE = 5.77%
Normalized Velocity MAE = 4.79%
Velocity R^2 = 0.9286

Observed Permeability = 1 m/day
Estimated Permeability = 1.108 m/day
Percentage Error = 10.8%

Task 3: Recap

- **First Weights:** Loaded pre-trained weights from Task 2 so the model begins training with a physically consistent pressure field.
- **Positivity-Enforced k:** Defined hydraulic conductivity k as a learnable scalar (`nn.Parameter`), constrained via softplus(k_{raw}) to guarantee $k > 0$.
- **Joint Loss Minimization:** Optimized k and pressure head H simultaneously by combining governing physics ($\mathcal{L}_{\text{PDE}}, \mathcal{L}_{\text{BC}}$) with velocity data loss ($\mathcal{L}_v$).
- **Optimization:** Used **Adam** for stable convergence.

Outlook & Discussion

Key Takeaways

- **Nondimensionalization:** Stabilized network gradients and eliminated scale imbalances.
- **Forward Model:** Predicted pressure head H purely from physics and boundary conditions.
- **Inverse Model:** Successfully learned conductivity k from velocity magnitudes while ensuring k>0 via `softplus`.
- **Adam + L-BFGS Strategy:** Adam provided rapid global convergence; L-BFGS fine-tuned boundary residuals to high precision.

Future Work & Outlook

- **Parametric PINN Model (Task 4):**
  - Include reservoir height h1 as a direct input variable: (x^,y^,h1)→H.
  - Train **one universal network** that predicts seepage for *any* water level instantly without retraining.
- **Improving Accuracy & Generalization:**
  - **Model Comparison:** Benchmark the model on unseen reservoir heights.
  - **Noise Robustness:** Evaluate how reliably k is learned when velocity measurements contain real-world noise.

This is us

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Getting Better Results

Vielen Dank
für Ihre Aufmerksamkeit

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